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RESEARCH ARTICLE

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Experimental analysis on the microstructural, mechanical, thermal and tribological properties of graphene nanoplatelets and molybdenum disulfide filled polyamide-6,6 novel hybrid composite

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Abstract

Novel hybrid composite based on polyamide-6,6 (PA-6,6) was developed with graphene nanoplatelets (GnPs) and molybdenum disulfide (MoS_2) as hybrid fillers reinforcement using melt-mixing and injection molding techniques. The structural, thermal, mechanical and tribological characteristics of the hybrid composite were assessed by Fourier transform infrared spectroscopy (FTIR), X-ray diffraction (XRD), thermogravimetric analysis (TGA), differential calorimetry analysis (DSC), tensile test as well as the friction and wear test, respectively. Compared to pure PA-6,6, the PA-6,6/GnPs/MoS_2 hybrid composites exhibited enhanced thermal stability. Notably, the composites demonstrated significant improvements in tensile modulus and strength, particularly at a 5 wt.% of hybrid filler loading, showing an increase of ~59% and ~21%, respectively. However, the impact strength exhibited an opposite effect on hybrid filler reinforcements. Furthermore, the fractured surface morphology was analyzed using FESEM images. Moreover, the tribological performance was assessed with a linear reciprocating tribological (LRT) test using ball-on-plate tribo-pair at different abrading frequencies (5 and 25 Hz) and normal loads (10, 30 and 50 N), revealing significant improvements in friction coefficient (CoF) and specific wear rate (W_{SR}) characteristics for the hybrid composite up to 5 wt.% of hybrid fillers addition, attributed to the abrasive and adhesive wear mechanism. The worn surface morphology was further investigated using the FESEM micrographs to better comprehend the wear mechanisms. Therefore, the optimal performance of the developed hybrid composite was observed at 5 wt.% of hybrid filler loadings. Therefore, the developed hybrid composites' excellent thermal, mechanical, and tribological properties could be suggested for their potential utilization in lightweight automotive applications.

KEYWORDS

dry sliding wear, friction and wear, GnPs hybrid composite, hybrid composite material, MoS_2 hybrid composite, PA-6,6 hybrid composite, tribology of hybrid composite

1 | INTRODUCTION

Polyamides, particularly polyamide 6 and polyamide 66, stand out as the most preferred category of engineering materials. They are highly recommended for applications such as bearings, gears, and cams due to their widespread availability, and they also possess a commendable blend of strength and tribological performance. The toughness, low friction coefficient, and excellent abrasion resistance of polyamides make them an ideal substitute for various materials, ranging from metal to rubber. Utilizing polyamide not only reduces the need for lubrication but also eliminates issues like galling, corrosion, etc., while simultaneously improving wear resistance and sound-dampening properties.[1] However, it is difficult to obtain the desired thermal and mechanical properties individually from the polymer matrix material. Polyamides contain various important shortcomings,[2] which can be further assessed by incorporating some modifications in polyamides for the improvement in their mechanical and thermal properties by reinforcing fillers, which have nowadays become an active research area. Therefore, the reinforcement of various micro/nano-scaled materials significantly improves the specific material properties that further open enormous research opportunities[3] in the domain of science and engineering. Reinforcing the micro/nano additives does not mean leading towards the desired thermal or mechanical properties, but the significant dispersion of additives/ fillers over the matrix phase with good filler-filler and filler matrix interaction is liable for their characteristic performance.[4] Such conditions may be achieved by various preparation techniques such as solution mixing, milt-mixing, or various other techniques that are considerably cost-effective. Apart from thermal and mechanical characteristics, lower friction and high wear resistance characteristics play a vital role towards the industrial application of such composites or hybrid composites due to their question of sustainability under stringent conditions of extreme pressure and temperature, high speeds, as well as abrading loads. The effect of nanofillers significantly contributed to determining the strength of composites.[5] Therefore, reinforcing filler materials with desired characteristics such as high thermal, mechanical, and lubricating behavior are good choices. In this context, graphene nanoplatelets (GnPs) and molybdenum disulfide (MoS_2) are widely studied materials with fascinating properties.

The enhancement of mechanical and other characteristics in these nanocomposites relies heavily on the properties of the nano/microparticles, including their shape, size, surface features, dispersion state, and filler loading percentages. It is worth noting that incorporating carbon-based fillers (nano/micro) such as carbon nanotubes, carbon nanofibers, and graphene nanosheets can not only substantially enhance mechanical, thermal, and electrical properties but also result in reduced processing costs. In this regard, GnPs reveal their capability to enhance the thermal stability of polymers by serving as barriers that impede the propagation of heat energy within the material.[6–8] Studies indicate that 2D platelet-like structure provides an efficiency of enhancing the mechanical properties as well as the thermal conductivity when compared to 1D rod-like structures such as carbon nanotubes (CNTs), carbon fibers, etc.[9] In this context, Zhang et al.[10] reported the improved tensile properties of graphene oxide (GO) loaded PA6 composite. The study reveals a significant improvement of about 88% and 66.5% in tensile strength and tensile modulus, respectively, just by an addition of 1 wt.% of GO filler contents. A similar study carried out by Mindivan[11] exhibited an increase of about 50% in tensile modulus in reinforcing the 4 wt.% of GnPs particles. In another sequence of studies, Mayoral and colleagues[14] observed a nucleating influence on PA6 crystallization resulting from the incorporation of GnPs through melt compounding. This process led to a significant enhancement in the crystallinity of the composites, reaching up to 120% at 20 wt.% GnPs loading. Notably, the young's modulus exhibited a remarkable 420% increase at 20 wt.% GnPs loading, aligning well with theoretical predictions. Furthermore, Lee and coworkers have also reported and studied the micromechanics of GnPs-reinforced thermoplastic composites.

In addition to mechanical properties, researchers have utilized various micro/nanofillers to improve the tribological properties of the composite materials. Graphene and its derivatives, copper/copper oxide/copper fluoride, graphite/graphene, diamond, carbon nanotubes, MoS_2, SiO_2, carbon nanotubes, silicon carbide, graphite fluoride, fluorographene, magnesium hydroxides, boric anhydride, boron nitride, aluminium oxide, halloysite nanotube, and titanium dioxide are few examples of fillers used by researchers to improve the tribological properties of polymers. The abrasive particles (carbides) were also found useful in previous research to produce a good wear-resisting characteristic, but the friction coefficient was found to be increased significantly. In this regard, solid lubricant materials such as graphite, MoS_2, silicon dioxide (SiO_2), diamond nano-powder, boric anhydride, and boron nitride. etc., play a crucial role in influencing friction and wear properties. Out of these examples, MoS_2 is the most commonly known solid lubricating material for providing better friction and wear behavior. MoS_2 is a dark blue/black crystallite sulfide of molybdenum that is a widely utilized two-dimensional solid lubricant material that contains a layer-wise sandwiched structure,[1,12,13] in which a molybdenum (Mo) layer is sandwiched between the other two sulfur (S) layers with a strong covalent bond between the Mo and S atoms and simultaneously possess weak van der Wall forces among the layers that provide an easy sliding of layers one over another.[14] Bulk MoS_2 is the most widely accepted 2-D material used for semiconductors with direct/ indirect band gaps along with strong photoluminescence characteristics.[15] These existing direct/ indirect band gaps directly depend upon the number of layers in MoS_2 material. Furthermore, MoS_2 material offers notable elasticity, robust tensile modulus, and resistance to elastic deformation,[16,17] so this can be introduced in a wide range of application areas such as science and engineering, electronics, medical, etc.

Many studies in the literature investigate the friction and wear behavior of polyamide and its composites. in this regard, Suresha et al.[18] and kalaitzidou and coworkers,[19] in their study, explored the potential of graphite material as a filler reinforcement in the reduction of CoF and W_{SR} properties of the polymer matrix composite along with the increased thermal and mechanical properties. In another research, Zhou et al.[20] have studied the effect of carbon fiber (CF) on polyphenylene sulphide (PPS) and PA-6 polymer blends for their tribomechanical properties. the study found a better wear behavior of PA6/PPS-CF composite for increased contents of carbon fiber under a high load of 20 N or a high-speed of 1500 rpm/min. Similarly, to enhance the tribological behavior, Masood and his coworkers[21] have investigated the GnPs-coupled polytetrafluoroethylene (PTFE) composite. The study found synergistically improved friction and wear behavior as well as adhesion to the substrate. The study observed an enhancement in friction and wear characteristics with an optimal graphene concentration of 0.5 wt.%. Bingli et al.[22] have investigated the tribological and mechanical behavior of monomer (MC) nylon/graphene oxide nanocomposites. The results showed that graphene oxide of very low content has a notable effect on reinforcing the MC nylon matrix. It was further found to have reduced wear and friction properties under dry sliding conditions. In another research by Guglani and Gupta,[23] the effect of micro-TiO_2 reinforced in PA-6,6 composite was studied for their tribomechanical properties. The study found that the composite exhibited reduced wear and friction characteristics of up to 6 wt.% of TiO_2 addition. However, 2 wt.% was found to be the most suitable reinforcement for improvement in tribological properties in pure PA-66. A similar study by Basavaraj et al. and his coworkers[24] investigated the influence of MoS_2 on the physicomechanical and tribological properties of PA-6,6 along with carbon black (CB). The investigation found a linearly increased tensile property along with increasing content of MoS_2 filer. The study also revealed that the introduction of MoS_2 in the presence of CB reduced the friction coefficient and wear behavior of PA-6,6, with improvement in laser etching resistance. Also, MoS_2 was potentially found to increase the adhesion between the transfer film and the counterface surface. Yinping et al.[25] reported that high loads and frequencies encourage the formation of a compact transfer film, which is considered the key mechanism behind the high load-bearing capacity and excellent wear resistance of bonded MoS_2 solid film lubricants.[25,26] In contrast, Liu et al.[27] noted that MoS_2 was not particularly effective in reducing friction and even increased the wear of PA 6. However, Steinbuch[28] observed that while MoS_2-filled polyamides could lower the wear rate, they did not reduce the friction coefficient. However, from the literature survey, it was observed that the pin-on-disk tribo-test was carried out extensively in tribology investigations. Very few work has been reported on pin-on-plate tribo-pair tribology testing. However, no study has been reported utilizing the ball-on-plate tribo-pair arrangement for tribology testing that includes the load and frequency parameters. Additionally, no study has been reported to date precisely on utilizing the GnPs and MoS_2 filler contents together in a hybrid filler mode loaded in equal weight percentage (wt.%) with PA-6,6 matrix. Also, very few works have been reported that investigate the tribological performance of the thermoplastics like PA-6,6, and no such study has been found that establishes the correlation of their structural, thermal, mechanical as well as tribological properties of this PA-6,6/GnPs/MoS2 (PGM) composites.

Therefore, the main objective of the present work is to explore the effect of hybrid filler contents (GnPs:MoS_2) in equal wt.% reinforcement in PA-6,6 matrix to investigate their structural, thermal, mechanical and tribological performances. The present work also focuses on the failure mechanism during tensile testing for this novel PGM hybrid composite using the fractured surface morphological study, which has rarely been found in the previous literature. Most importantly, the current investigation also includes the effect of transfer layer formation on the counterface surface and the tribo-chemical reactions between the MoS_2 and GnPs fillers on the tribological properties of the PA-6,6 matrix. Therefore, PA-6,6 was compounded with GnPs and MoS_2 fillers. In this work, the MoS_2 has been reinforced to produce the lubricating effect and is a kind of material that decomposes into MoO_3, FeS, $FeSO_4$ and $Fe_2(SO_4)_2$ during sliding action against the steel-ball counterface surface. These compounds could enhance the adhesion between the transfer film and the counterface surface. Normally, the matrix should withstand high temperatures so that it could sustain high frictional heat generation during the sliding action. To fulfill this requirement, an additional filler with high thermal conductivity offers a great advantage, especially when preventing the temperature increase in the contact area, which is crucial to avoid higher wear rates. GnP is one of the materials that generate a conductive pathway to dissipate heat generated during friction, which further improves the tensile and wear characteristics.

However, its tribological performance has not been explored much in terms of its integration with the PA-6,6 polymer matrix. Therefore, in the present work, both GnPs and MoS_2 have been reinforced together in equal wt.% to have a synergistic effect in transfer film formation, which would increase the wear resistance characteristics of the novel hybrid PGM composite. Under these circumstances, both the fillers have been reinforced in equal wt.% to develop the hybrid PGM composites. To investigate their friction and wear performance, the linear reciprocating tribology (LRT) test, utilizing the ball-on-plate tribo-pair arrangement, was carried out under low and high load and abrading frequency, which has not been explored much to date for such a novel hybrid composite.

2 | METHODOLOGIES AND EXPERIMENTAL DETAILS

2.1 | Materials

In the current research work, the commercially available Polyamide-6,6 (PA-66) (unfilled natural, PXR-01 NC), supplied by Next Polymers Ltd. (INDIA) under the brand name Celanese, having a density of 1.14 g/cm^3, was used as the matrix material. GnPs, having a density of 2.2 g/cm3 and a mean particle size of 5 μm, and the MoS_2, having a density of 5.06 g/cm3 with a mean particle size less than 2 μm were procured from Sigma-Aldrich Chemical Pvt. Ltd. (INDIA).

2.2 | Fabrication of fillers reinforced hybrid composite

The matrix and filler components were precisely weighed according to the prescribed proportions and processed using a Haake twin screw small extruder (Thermo Haake). The materials are dried in a vacuum oven (MCP, HEK—Germany) at 90°C for 6 hr to eliminate the moisture contents before the extrusion process. For different testing procedures, various samples were prepared via the melt-mixing technique[29,30] by reinforcing the filler materials in equal weight percentages (wt.%). The filler materials (GnPs and MoS_2 in equal wt.%, i.e., in 1:1 ratio) were efficiently premixed using a fine handheld blade in separate batches for various sample preparations. Initially, PA-6,6 granules were fed into an extruder and processed for 2 min to melt down the polymer. The extruder was then loaded with a preset amount of premixed GnPs and MoS_2 fillers and processed further for 3 min at a screw speed of 130 rpm with temperatures ranging from 275 to 285°C. The extruded melt was collected in a cylindrical gun having a temperature range of 300–325°C and then injected into a Haake mini-injection molding machine (Thermo Electron Corp., Mini jet) having an injection pressure of 600 bar and injection time of 10 s. This process produces hybrid composite samples of PA-66/GnPs/MoS_2 (PGM composite) by reinforcing the polymer with equal weight percentages of GnPs and MoS_2 particles for various testing purposes. The composition of the prepared composite samples, along with the necessary information on filler contents in wt.%, is provided in Table 1.

2.3 | Characterization technique

2.3.1 | Surface morphology

Moreover, the morphological observations of the prepared injection molded PGM hybrid composite were carried out to study the filler particle dispersion as well as the tensile fractured surface of composite samples to identify the various causes related to fracture of the sample with varying wt.% of fillers contests were observed using the Field Emission Scanning Electron Microscope (FE-SEM) (Sigma 300, Carl Zeiss Microscope Ltd., Germany) in high vacuum at the voltage of 20 kV.

2.3.2 | Microstructural study

The Elemental compositional analysis of the hybrid composite samples was carried out using an energy-dispersive spectroscopy (EDS) attachment connected to the FE-SEM facility. The EDS analysis was carried out using the EDAX (Ametek). In order to make the observed portions appropriately conductive, the samples were undergone for a preliminary metallization with a gold coating.

2.3.3 | Fourier transform infrared radiation

A Perkin Elmer 100 spectrometer with an ATR sampling adapter was utilized to identify the presence of various functional groups in the developed PGM hybrid composite with Fourier Transform Infrared Radiation (FTIR). All spectral measurements were obtained at a resolution of 4 cm^{-1} with 32 scans in the 4000–400 cm^{-1} under mid-infrared light band.

2.3.4 | Study of x-ray diffraction (XRD) analysis

A high-resolution x-ray diffractometer (Rigaku smart lab, 9 kV) was utilized for XRD characterization of the prepared

TABLE 1 Composition of the prepared PGM composite materials.

Sample code	PA-6,6 (wt.%)	GnPs: MoS_2 (wt.%)	GnPs (wt.%)	MoS_2 (wt.%)
Pure PA-6,6	100	0	0.00	0.00
PGM-1%	99	1	0.5	0.5
PGM-3%	97	3	1.5	1.5
PGM-5%	95	5	2.5	2.5
PGM-10%	90	10	5	5

hybrid samples using Cu-Kα radiation ($\lambda = 1.54184$ nm). The hybrid composite samples were scanned at 3° min^{-1} with a scanning rate of $2\theta = 1$ to 80°.

2.4 | Thermal characterization

2.4.1 | Study of differential scanning calorimetry

The DSC tests were performed on samples weighing 3–5 mg sealed in an aluminium crucible using the Discovery DSC 25 (Waters, USA) by employing the heat-cool-heat scanning method to estimate the melting and crystallization behavior of the PGM hybrid composite at a scanning rate of 10°C/min, under the inert nitrogen condition, following a standard step-by-step test protocol that included heating the samples from 25°C (room temperature) to 300°C, isothermal stasis A steady thermal rate of 10°C/min was maintained throughout all non-isothermal stages. Furthermore, DSC thermogram processing is mostly confined to the second heating in order to remove any earlier thermomechanical history of thermal processing, which was also required to stabilize the material. The heat was calculated by taking the integrals of the onset points of the relevant peaks and the areas under the curves. Using the relation stated in Equation (1), the average values of the enthalpy of fusion (H_f) were utilized to compute the percentage of crystallinity (X_c).

$$X_c(\%) = \left(\frac{\Delta H_f}{(1 - W_f)\Delta H_m^0}\right) \times 100 \tag{1}$$

Here, X_c shows the percentage crystallinity, H_f indicates the enthalpy of fusion, and ΔH_m^0 represents the enthalpy of melting of 100% crystalline material (195 ± J/g for PA66).[31]

2.4.2 | Study of thermogravimetric analysis

Furthermore, it is a well-known fact that the thermal decomposition of polymeric material is due to the rise in temperature to the threshold, which produces the extreme vibrational energy that causes the rupture of bond links among the polymeric chains. This states the importance of the thermal stability of polymeric materials, which is a function of bond energy.[32] In this regard, for a more in-depth and better understanding of the thermal stability characteristics of the hybrid composite samples, the thermo gravimetric analysis (TGA) was performed using a DTG-60 (Shimadzu Corporation, Japan) TGA/DTA analyzer, which allows the recording of the weight loss data of the substance as a function of temperature. The experimental analysis was carried out in a platinum pan with samples (approximately 3–5 mg) in an inert nitrogen atmosphere having a nitrogen flowing rate of 50 mL/min. The samples were then heated from 30 to 600 °C at a rate of 10°C/min.

2.5 | Mechanical properties characterization

2.5.1 | Tensile properties

For tensile and flexural property assessment of the prepared PGM hybrid composite, the tensile test was carried out as per the ASTM D638 by using the Instron-3366 universal testing machine (UTM) equipped with a dynamometer-based load cell having a capacity of 30 kN. The cross-head speed during the tensile test was maintained to be 5 mm/ min. Furthermore, under the same experimental setup with an additional attachment of a three-point bending test, the flexural characteristics of the procured hybrid composite samples were investigated as per the ATM D790.

2.5.2 | Impact strength

At room temperature, Izod impact tests were performed utilizing an Izod impact tester IT504 (Tinius Olsen, INDIA) impact tester with an 11.224 J hammer in accordance with ASTM D256. Three samples for each composition of hybrid PGM composite were examined in order to determine their impact strength under notch opening mode (tension mode).

2.6 | Tribological performance

The ball-on-plate wear testing tribometer (TR-705, DUCOM, India) was employed to investigate the tribological characteristics of the PGM hybrid polymeric composite as per the ASTM G133-05 standards, which outlines the procedures for performing linearly reciprocating ball-on-flat sliding wear tests.[33]

The wear test of the specimens was performed under the dry sliding condition, and their weight change was measured in an electronic weighing balance (Mettler Toledo, USA), with a sensitivity of 0.0001 g before and after the wear test. The steel ball counterface material used was of high chromium steel (100Cr6) with chemical compositions of (1.05% C, 0.35% Si, 0.50% Mn, 0.025% P, 0.015% S and 1.60% Cr) and hardness of approximately 60–66 HRC with a highly polished surface with an average surface roughness (R_a) of ~0.024 μm. Furthermore, to ensure uniformity, all the polymeric samples underwent the fine cloth polishing procedure utilizing the polishing machine Struers (LaboPol-5) Japan, equipped with a rotating disk with a velvet polishing cloth, leading to a uniform surface finish achieved consistently across all the samples.

The wear tests were separated into two categories, that is, 5 and 25 Hz frequency levels. In these two categories, the wear test was performed by employing normal loads of 10, 30 and 50 N to examine the wear behavior of the composite samples. The stroke length and run-in-time were kept constant at 5 mm and 30 min, respectively, for each sample during the test. Prior to the testing, the samples underwent a cleaning process to remove any dirt or foreign materials on the surface and were subsequently dried at room temperature. The steel ball was securely affixed to the ball holder, where the accuracy of position repeatability is 1 μm, ensuring the ball-holder perpendicularity to the flat plate sample upon contact, thereby fulfilling the essential conditions for proper contact. The tests were conducted continuously without interruptions at a room temperature of 25 ± 2°C. After each test, the specimen was carefully removed, cleaned of any wear debris using ethyl alcohol, and dried before further analysis.

The mass losses of the specimens were determined according to Equation (2), where m_i is the initial mass taken before the wear test, m_f indicates the final mass of the specimen after the completion of the wear test, and Δm denotes the mass loss of the specimen.

$$\Delta m = \left(m_i - m_f\right) \tag{2}$$

To evaluate the material density, the ASTM D792 standard was followed for determining the density of plastics,[34] using a density test kit. Five tests were conducted on each sample for all compositions, and the average values were documented. Moreover, to evaluate the wear volume (ΔV), the mass loss was divided by the calculated density of the materials, as indicated in Equation (3).

$$Volume\,loss\,(\Delta V) = \left(\frac{MassLoss,(g)}{Density,(g/cc)}\right) \times 1000 \rightarrow \left(mm^3\right) \tag{3}$$

Further, the data derived from the wear test enables the evaluation of the specific wear rate (W_{SR}) by employing Equation (3) as follows:

$$W_{SR} = \Delta V / F_N L \left(mm^3/Nm\right) \tag{4}$$

Where, ΔV denotes the wear volume in mm^3, F_N indicates the applied normal load in Newtons (N), and L is the stroke length in meters (m).

3 | RESULTS AND DISCUSSION

3.1 | Morphological observations

Figure 1 illustrates the morphological observations of the developed PGM composite. The FESEM images delineated in Figure 1A,B demonstrates the stacked layers of GnPs and MoS_2 filler contents. Figure 1C exhibits the significant dispersion of GnPs and MoS_2 filler particles over the entire region of the PA-6,6 matrix phase. The sheet-like structure of GnPs and MoS_2 and their loading on the pure PA-6,6 can be clearly seen from the FESEM image. EDS spectrum analysis further confirmed the reinforcement of GnPs and MoS_2 fillers. Figure 1D exhibits the developed peaks of Carbon (C), Oxygen (O), Molybdenum (Mo), and Sulfur (S) particles that are clearly visible from the EDS spectrum analysis. Also, the presence and distribution of these hybrid filler contents in the PGM hybrid composite are further confirmed through the EDS mapping, as shown in Figure 1E. From this morphological study, it can be clearly stated that the developed PGM hybrid composite is loaded with GnPs and MoS_2 hybrid filler contents.

3.2 | FTIR spectroscopy

Figure 2 depicts the FTIR spectra of PA-66 and developed PGM hybrid composite with 1, 3, 5, and 10 wt.% of hybrid fillers (GNP:MoS_2) in the ratio of 1:1 addition. The spectral range at 3316–3298 cm^{-1}, present at Pure PA-66 and

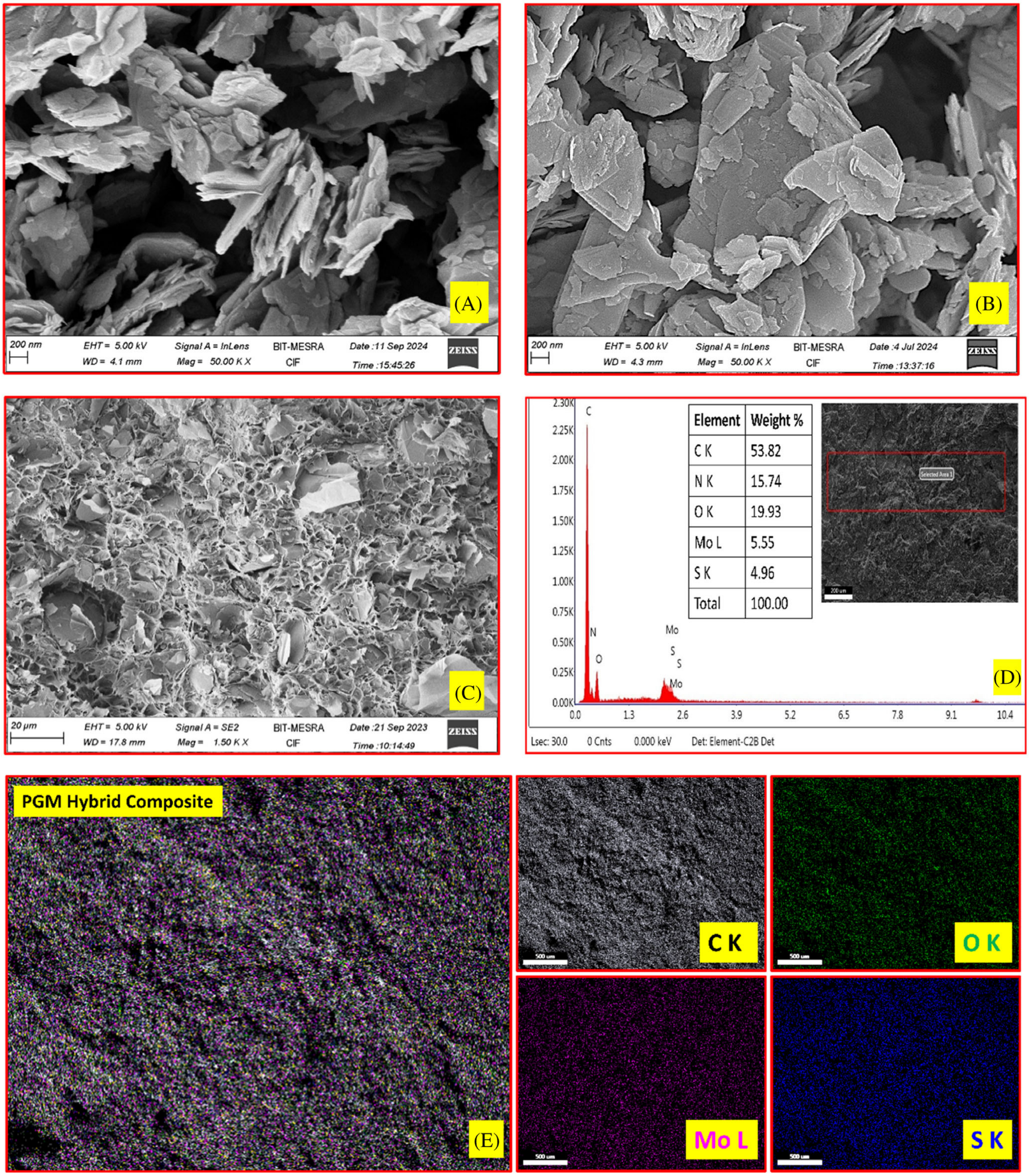

FIGURE 1 FESEM images showing (A) GnPs particles; (B) MoS_2 particles; (C) dispersion of hybrid fillers (GnPs:MoS_2) in PA-6,6 matrix; (D) EDS spectrum analysis of PGM hybrid composite with 10 wt.% of hybrid filler loadings; (E) elemental distribution in PGM hybrid composite with 10 wt.% of hybrid filler loadings.

its PGM hybrid composite samples, is usually assigned to N-H stretching and bending vibration occurring in PA-6,6.[35] The spectral peak at 3316 cm^{-1} displayed the broader peak of PA-6,6 polymer due to the H-bonded hydroxyl (—O—H) group. This broader peak further moves towards the lower wavenumber with reinforcement of MoS_2 particles in the hybrid composite.[36] The spectral peak developed at 3074 cm^{-1} (PGM 1, 3, 5, and 10 wt.%)

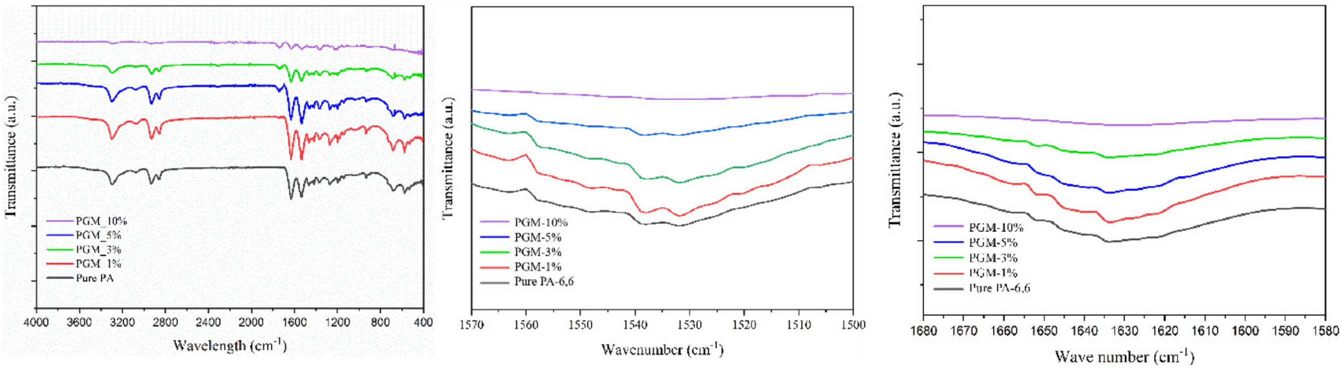

FIGURE 2 FTIR spectra analysis: (A) PA-6,6,/GnPs:MoS_2 hybrid composite, (B) enlarged view of spectral peaks showing N—H bonding, and (C) C—O bonding.

exhibits the characteristic bands of the O—H group.[37] The peak identified at 2931 cm^{-1} is due to the —CH_2 asymmetry of the methyl group, and again, the spectral peak observed at 2872.52 cm^{-1} is due to the —CH_2 symmetry stretching of the methylene group,[32] and the spectral range from 2927 to 2854 cm^{-1} showing the —CH stretching of the ethylene sequence.[38] The spectral peak detected at 1531 cm^{-1} is due to the bending vibration of the N—H group in PA-66.

Additionally, from Figure 2A,B, the absorbance spectral peak ranges from 1633 to 1645 cm^{-1} associated with C═O amide-I stretching, and the characteristics spectral ranges from 1529 to 1551 cm^{-1} attributed to the combination of C—H and N——H amide-II stretch are both discernible.[39] Moreover, the spectral ranges from 1402 to 1463 cm^{-1}, 933 cm^{-1}, and 683 cm^{-1} are attributed to the MoS_2. The band ranges from 505 to 443 cm^{-1}.[40] Further, at the spectral wavelength of 933 cm^{-1}[37] and 683 cm^{-1}, the absorbance peak of the S—S bond and Mo—O was observed.[41]

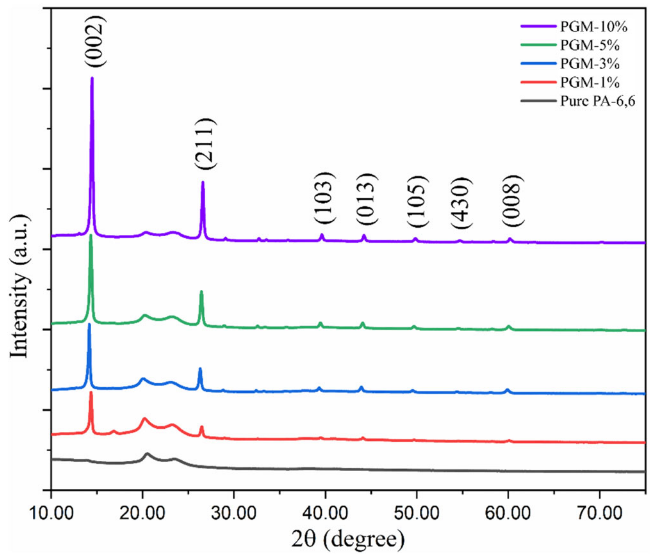

FIGURE 3 XRD pattern of PGM hybrid composite obtained from different concentrations of hybrid fillers.

3.3 | XRD analysis

XRD analysis was carried out in order to evaluate the crystalline phase and dispersion of GnPs and MoS_2 filler contents in the PA-6,6 matrix phase. Figure 3 delineates the XRD pattern obtained for the PA-6,6/GnPs:MoS_2 (PGM) hybrid composite system. The given stacking plot in the figure shows that the peak positions and corresponding peak intensities exhibit the crystalline phase of GnPs and MoS_2 filler particles reinforced with PA-6,6 matrix material. The amorphous phase of the pure PA-6,6 polymer matrix can be identified by major broader peaks obtained at $2\theta-$values of 20.489° and 23.593° in the XRD pattern, suggesting that the presence of crystals in the PA-6,6 matrix phase is very small in size.[42]

Further, compared to the pure PA-6,6 polymer matrix, the reinforcement of GNP shows very sharp peaks that have been identified at $2\theta-$values of 26.44° (d = 3.364 Å) with crystal plane (211). The corresponding peak of GnPs particles is also one of the characteristic peaks of pure GnPs that completely match the standard JCPDS database with card no. 00-03-0401. The increasing wt.% of GnPs increases the peak intensities. The formation of such characteristic peaks at the corresponding $2\theta-$value suggests the presence of GnPs particles in the PA-66 matrix phase. Other smaller peaks at $2\theta-$values of 44.07° (d = 2.049 Å) and 54.47° (d = 1.693 Å), corresponding to the crystal planes (013) and (430), respectively, indicate the uniform dispersion of GNP particles in the matrix phase. Similarly, the hybridizing effect with MoS_2 filler addition shows a significant sharp peak $2\theta-$values of 14.37° (002) with a d-spacing of 6.155 Å. This indicates a well-stacked layered structure of MoS_2 presence in PGM hybrid composite.[43] Similarly, the other addition peaks of MoS_2 in PGM hybrid composite due to 1, 3, 5, and 10 wt.% of GNP:MoS_2 addition in a 1:1 ratio, can be clearly observed at 39.54° (d = 2.277 Å), 49.78° (d = 1.829 Å), and 60.14 (d = 1.537 Å), corresponding to (103), (105), and (008) crystal planes, respectively. This indicates a good agreement with the previously available data and JCPDS database card no. 00-037-1492. Therefore, it has been found that sharper peaks were observed at 10 wt.% of hybrid filler reinforcement in PGM-10% hybrid composite compared to the pure PA-6,6. This indicates that the PGM hybrid composite with 10 wt.% hybrid filler contents demonstrates higher crystallinity compared to pure PA-6,6, which further provides an indication of higher mechanical strength. Thus, the XRD results also support the mechanical properties of the composite materials.[44,45]

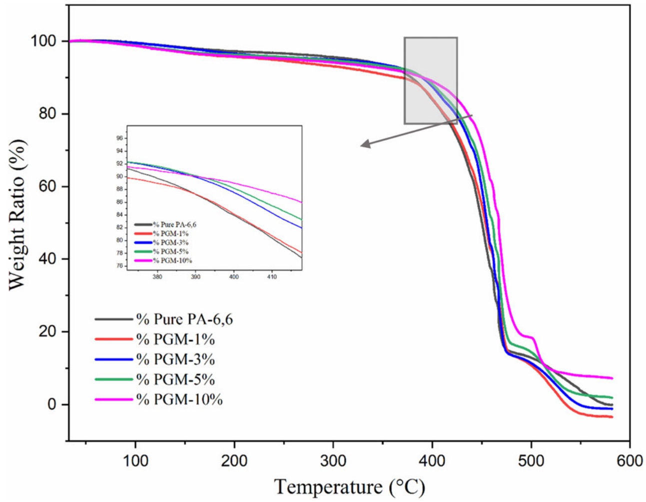

FIGURE 4 TGA curves for PGM hybrid composite developed using the hybrid fillers GNP:MoS_2 (1:1 ratio).

3.4 | Thermal properties

3.4.1 | Thermogravimetry analysis (TGA)

TABLE 2 Thermal data obtained from the TGA analysis for PGM hybrid composite.

Sample code	T_{10} (°C)	T_{50} (°C)	T_{99} (°C)
Pure PA-6,6	357.64	450.07	566.77
PGM-1%	367.17	448.37	539.49
PGM-3%	388.50	455.08	544.85
PGM-5%	387.14	460.90	566.46
PGM-10%	389.54	566.46	584.54

The TGA thermogram of the developed PGM composite system is delineated in Figure 4, and the corresponding thermal stability parameters are tabulated in Table 2. The TGA analysis was carried out basically to understand the influence of the hybrid fillers on the thermal behavior of the PGM hybrid composite. The thermal stability of polymer composite system depends mostly upon their morphological factors, the crystallinity of the available matrix phase, filler size, filler matrix interfacial adhesion, as well as the degree of dispersion of fillers over the matrix phase. In the present work, the thermal stability of the PGM composite system is analyzed under the nitrogen atmospheric condition. From Figure 4, it can be clearly seen that the thermal stability of the prepared hybrid composite system (PGM) is improved significantly due to the reinforcement of hybrid fillers in the PA-6,6 polymer matrix. The thermal breakdown temperature for pure PA-6,6 and hybrid composite was found to be in the range of 357.64–589.54°C. The 10 (T_{10}), 50 (T_{50}), and 99 (T_{99}) % of weight loss due to the thermal decomposition of pure PA-6,6 was found to be at temperatures 357.64, 450.07, and 566.77°C, respectively. At increasing wt. % of GNP and MoS_2 (1, 3, 5 and 10 wt.%) in hybrid filling mode in equal amounts (1:1 ratio) increases the thermal stability. For 1% of GNP:MoS_2 reinforcement, no significant change at 50% thermal decomposition is identified. This indicates that 1% of GNP:MoS_2 hybrid loading does not reach the percolation threshold level for the said hybrid composite system.[46] At 10 wt.% of filler contents (GNP:MoS_2) in PGM hybrid composite, the thermal breakdown for T_{10}, T_{50}, and T_{99} was identified at temperatures 389.54, 566.46, and 583.54°C, respectively. A continuous increase in thermal breakdown for 10, 50 and 90% weight loss was noticed significantly. At the maximum of 10 wt.% of GNP:MoS_2 addition shows an enhancement in thermal stability by 31.9, 116.39, and 16.77°C, which are 8.919%, 25.861%, and 3.135% higher than the pure PA-66 corresponding to the 10, 50, and 99% weight loss of the hybrid composite system. This increase in thermal stability is due to the development of a protective layer due to hybrid filler addition over the matrix phase, which also suggests a proper mixing and good matrix filler interfacial adhesion. The developed protective layer produces a barrier effect that hinders the transportation of temperature up to the matrix phase and protects against thermal degradation. The physical barriers created by the hybrid fillers reduce the transit of volatile breakdown of products into the composite system during the thermal decomposition process after the PGM hybrid filler addition to the matrix phase.[47] As a result, while at a greater temperature, the PGM hybrid composites show the same level of deterioration as the PA-6,6 matrix. Similar studies have also been carried out previously that support the current findings. In this regard, Shueb et al.[24] carried out an investigation to show a slight improvement in the thermal stability of the PA-6,6/GnPs composite. The study reveals an increase in the onset thermal degradation temperature by 10 °C. Russo and co-workers also identified a similar effect.[3]

Furthermore, the thermal properties of GnPs reinforced polycarbonate (virgin and recycled) composite were also carried out by Wijerathne et al.[48] The study reflected that the incorporation of GnPs significantly affects the thermal stability and glass transition temperature (Tg). The study showed that the GnPs reinforcement into the PC matrix had only a little impact on thermal stability and glassy transition temperature (Tg). Thermal stability was best for virgin PC/GnPs composites at 1 wt.% (2.74% increase compared to virgin PC) and at 10 wt.% (2.42% increase compared to recycled PC) GNP loading. Similarly, Altavilla and coworkers[49] demonstrated an elevated thermal stability of PS/MoS_2@oleylamine composite at a 0.3% concentration of MoS_2@oleylamine. Similarly, Zhou et al.[50] reported thermally stable behavior of the modified MoS_2/PS composite with admirable reinforcing ability, fire retardancy, thermomechanical properties as well as thermal stability. Such improved thermal results were observed due to the barrier effect and better compatibility of the polymer matrix with organic and inorganic filler materials, which formulates a strong interfacial attraction between the filler and polymer matrix.[12] Therefore, the present findings in PA-6,6 hybrid composite were found to prove the admirable thermal stability properties of GnPs:MoS_2 filled hybrid composite. Additionally, the improved thermal stability is attributed to the enhanced mechanical properties of the composites, such as mechanical strength, toughness and elasticity. High thermal stability due to the reinforcement of various fillers helps in minimizing the softening effect of the polymers, facilitating an efficient heat dissipation during the wear phenomenon and could stabilize the friction coefficient and wear rate at higher temperatures.[51,52] This reduces severe adhesive wear characteristics.

3.4.2 | Differential scanning calorimetry (DSC)

In addition to the XRD, DSC is also used for analyzing the crystallinity of the polymer. The melting enthalpy of the polymer is proportional to its crystallinity. High crystallinity corresponded to the increased melting enthalpy.[53] Figure 5 delineates the DSC melting and crystallization curve for PGM hybrid composites with varying wt.% of hybrid filler loadings in equal wt. the ratio of filler addition. The DSC curves show the heat-cool-heat method of the DSC scan. The DSC thermograms of the PGM hybrid composite were examined after erasing the thermal history in the 1st heating cycle under the N_2 atmosphere. The melting temperature (T_m), enthalpy of fusion (ΔH_m), crystallization temperature (T_c), enthalpy of crystallization (ΔH_c) and degree of crystallization were obtained from these DSC scans and have been tabulated in Table 3.

The DSC scan of pure PA-6,6 was found to have two distinct melting peaks, as depicted in Figure 5A. It is widely recognized that polyamide possesses two crystal forms: α-crystalline regions characterized by hydrogen bonds between antiparallel chains and γ-crystalline regions characterized by hydrogen bonds between parallel chains indicated by T_m peaks at 252.82 and 261.78°C. The MoS_2 filler addition also promotes the formation of γ-crystal in the PA-66 matrix, and the proportion of γ-crystal also increases by increasing the MoS_2 filler contents in the matrix phase. Further, the GnPs filler loadings play a very significant role in the PA-6,6 polymer matrix. From Table 3, it can be clearly identified that up to 5 wt.% of the hybrid filler loadings, the crystallinity was found to be increased by ∼7% than pure PA-6,6. Such an increase in crystallinity is mostly due to the incorporation of MoS_2 contents, which also provides insight into the enhanced mechanical properties such as increased stiffness, strength and toughness. At higher loading of the hybrid filler (10 wt.%), the crystallinity was further found to be decreased due to the depressed crystallization behavior of PA-6,6 polymeric chains.[54] Additionally, crystallinity affects the wear resistance property of the composite. More crystalline regions in the polymer tend to provide better wear resistance due to increased hardness and less molecular mobility during sliding. It is worth noting here that the GnPs filler addition does not alter much the melting temperature, but the presence of MoS_2 influences the melting temperature of the composite and produces high thermal stability of the hybrid PGM composite.[55] Moreover, the enthalpy of melting (ΔH_m) was found to be increased up to 5 wt.% of GNPs loading than PA-6,6 and decreased further with the addition of 10 wt.% of GnPs loading. This has a significant proportional relationship to the overall crystallization of the polymer.

Further, the study of cooling thermograms, as depicted in Figure 5B, clearly indicates a significant increase in the crystallization temperature. Specifically, the T_c of the PGM composite containing the 10 wt.% of hybrid filler loading gives a remarkable enhancement of 247.37°C compared to the pure PA-6,6 matrix. Such an increase in the T_c is due to the hybridized effect of GnPs and MoS_2 filler loadings in equal wt. ratio. Moreover, the MoS_2 filler loading hybridized with GnPs filler contents, suggesting that GnPs:MoS_2 filler loading in a hybridized manner acted as a nucleating agent for the PA-66 matrix.[30,32,56]

The incorporation of hybrid filler produces a combined hybridized effect, resulting in a heterogeneous

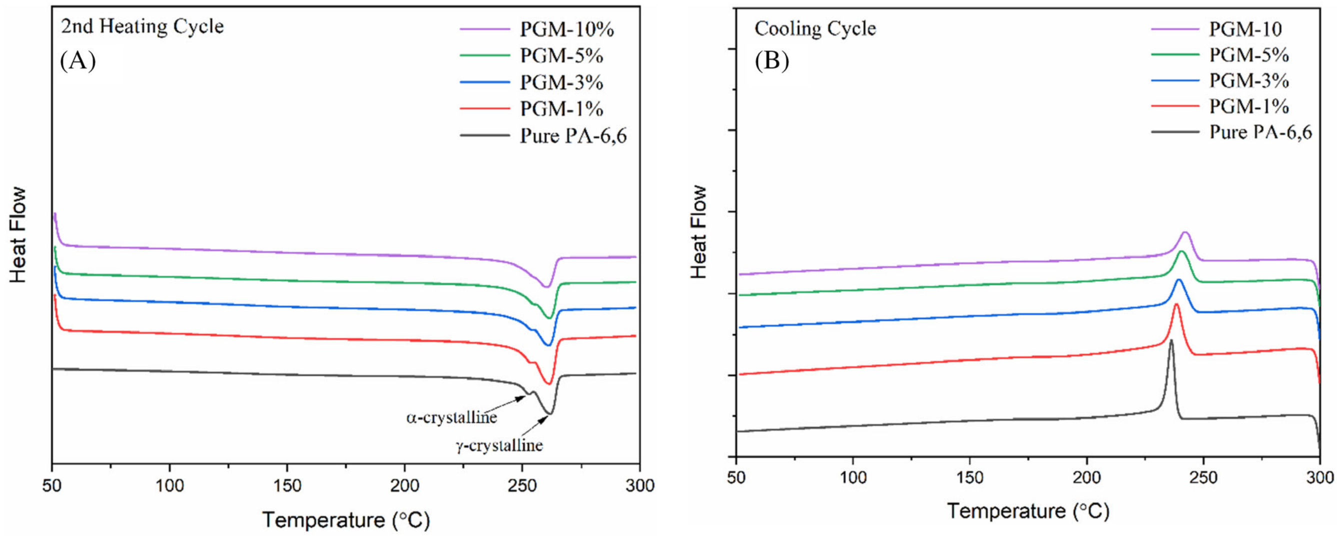

FIGURE 5 DSC thermograms indicating (A) second heating cycle and (B) cooling cycle of pure PA-6,6 and its PGM composite.

TABLE 3 The DSC scan data for the second heating cycle.

Sample code	T_m	ΔH_m	T_c	ΔH_c	X_c (%)
Pure PA-6,6	252.38	62.703	238.74	47.515	26.74
PGM-1%	252.01	68.873	243.18	50.914	30.47
PGM-3%	250.73	64.286	245.80	40.281	28.45
PGM-5%	250.98	64.549	246.63	35.366	28.56
PGM-10%	248.38	54.208	247.37	33.521	23.99

nucleation during the crystallization process of the composite. This is attributed to its large specific surface area, which reduces the energy needed to form new crystal planes and enhances the crystallization rate.[57] Additionally, the presence of MoS_2 filler particles reinforces the density of crystal nuclei in the system, improving the composite's crystallization rate. This underscores the role of MoS_2 as a wear-resisting agent, facilitating heterogeneous nucleation and thereby promoting easier nucleation of nylon molecular chain segments within the composite material.[56]

3.5 | Mechanical properties

The tensile test of the developed PGM hybrid composite was studied to investigate the influence of hybrid filler reinforcement in equal amounts. The study of mechanical characteristics is an important factor in understanding the characterization of fractured surfaces that helps in estimating the filler-matrix interaction,[53] filler adhesion, as well as the distribution and orientation of filler particles over the matrix phase.[55]

In the present work, the MoS_2 has been reinforced in equal wt.% with GnPs filler particles to investigate the mechanical characteristics such as tensile modulus, tensile strength and strain at break, as well as the fractured surface morphology. The present work is an extension of previous work on the PA-6,6/GnPs composite system.[55] Figure 6A–C delineates the tensile properties of the PGM hybrid composite. Figure 6A demonstrates the tensile modulus of a prepared hybrid composite of different wt.% of hybrid filler reinforcement. A significant increasing trend can be clearly identified in the tensile modulus characteristics on varying wt.% of filler reinforcement. The systematic reinforcement of hybrid fillers (GnPs:MoS_2) at different compositions with equal wt.% improves the tensile modulus of the prepared hybrid composites significantly. It was observed that at 1, 3, 5 and 10 wt.% of hybrid fillers reinforcement, the tensile modulus is increased by ∼44%, ∼52%, ∼59%, and ∼90% than pure PA-6,6, respectively. This result may be due to the fact that GnPs reinforcement provides the stiffening effect to the composite sample,[55] and the reinforcement of MoS_2 helps in providing a significant interaction of MoS_2 particles with the PA-6,6 matrix due to its lubricating property. This is due to the presence of solid interactions like H-bonding between the MoS_2 particle and the PA-6,6 matrix.[36] The polymeric chains attached to the MoS_2 surface, due to H-bonding, generate the nano-confined region that effectively improves the volume fraction of the composite.[58] The increased volume fraction facilitates the transmission of stresses from the matrix to the MoS_2 filler nanosheets.[36] In the present work, the developed hybrid composite seems to reflect a much more significant result than the previous work on the PA-6,6/GnPs composite.[55] This suggests the path for further hybrid composite development.

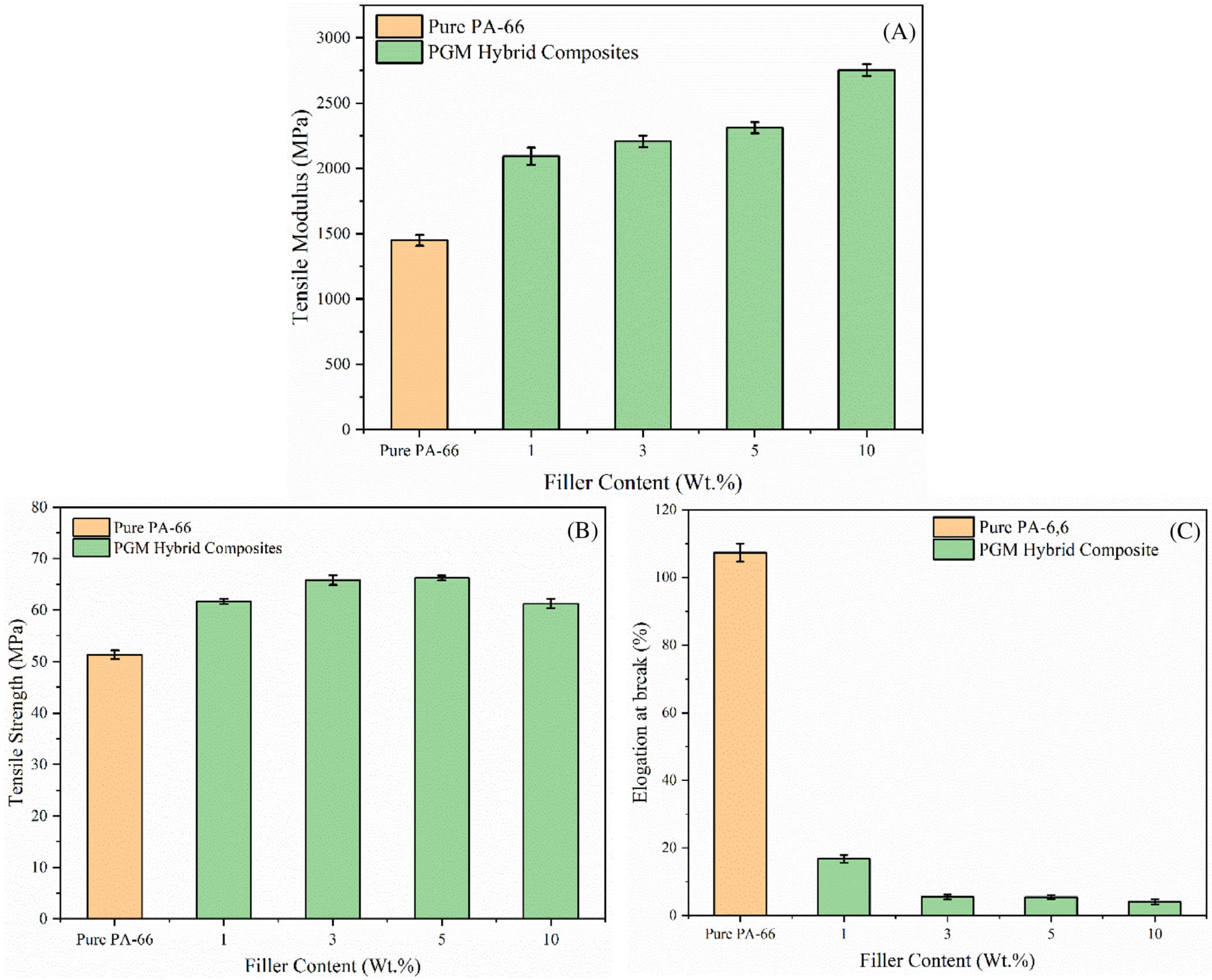

FIGURE 6 The mechanical properties of the PGM hybrid composites show (A) tensile modulus, (B) tensile strength, and (C) elongation at break characteristics.

Similarly, the tensile strength of the PGM hybrid composite was found to increase by up to 5 wt.% of the hybrid fillers addition, as shown in Figure 6B. The tensile strength seems to be increased from ∼12% to ∼21% at 1 to 5 wt.% of hybrid fillers loading. Such enhancement in tensile strength is due to better filler-matrix dispersion, decent filler-matrix interfacial adhesion, good retention of hybrid filler particles, and better stress transfer characteristics.[55] The significant hybrid filler dispersion and its interaction with the matrix phase can be clearly seen in Figure 1A–C. The GnPs addition hinders the chain mobility and provides a much stiffer structure.[59] Even a small wt.% of fillers have the tendency to scatter in a larger area where it can combine with the matrix material and can experience an effective improvement in stress transmission. Also, the GnPs content exhibits a significant contribution to the improvement in tensile strength due to the good compatibility and strong binding forces with the PA-6,6 matrix.[60] It is worth noting that the MoS_2 also produces a positive effect in the improvement of tensile strength, but at the same time, the excessive reinforcement may produce an opposite effect. The present work also reveals that with the excessive reinforcement of hybrid fillers (at 10 wt.% of GnPs:MoS_2), the tensile strength was found to be decreased, as shown in Figure 6B. This may happen due to the excessive reinforcement of hybrid filler particles where the GnPs play a dominant role in the composite behavior. The excessive GnPs and MoS_2 filler content is attributed to the factor of stress concentration caused by agglomeration.[61] Higher loading of MoS_2 may also lead to the formation of defects and nanochannels due to uneven and inefficient wrapping of filler content with the matrix phase. As a result, the deterioration in mechanical properties is observed.[62]

This leads to a sudden decrease in tensile strength. Also, the shear slip occurs under stress development in the composites due to having the lamellar structure of MoS_2.[63] This result shows a good agreement with the previous literature reported by Mittal et al.[64] and Puértolas et al.[65] The present work of PGM hybrid composite exhibits a significant enhancement in tensile strength compared to the previous work of PA-6,6/GnPs composite.[55]

Generally, in the case of polymeric materials, the incorporation of inorganic micro or nanofillers signifies the enhancement in tensile characteristics of the polymer composite, but as a common trend, the reduction in elongation at break is also observed.[36,66] Such reduction in elongation at break occurs by sacrificing the ductile nature of polymer composite along with the increasing wt.% of filler contents.[67] Since elongation at break is an important characteristic for identifying and analyzing the ductile behavior of composite. Furthermore, the improvement in tensile strength by reinforcing various fillers also imparts brittleness characteristics to the composite system. Figure 6C indicates a significant reduction in elongation at break characteristic of the developed PGM hybrid composite. At such higher reinforcement (10 wt.%) of hybrid fillers, the maximum reduction in strain at break was observed. Such significant reduction may be due to the vulnerable effect of GnPs agglomeration at excessive loading. Furthermore, the Van der Waals interaction between the GnPs nanosheets diminishes the GnPs/PA-6,6 interaction and hinders the effective stress transfer.[68–70] Meanwhile, the elongation at break of the composite system decreases. Similar results also agreed with other works found in the literature reported by Suresha et al.[67] and Lan et al.[68]

3.6 | Fractured surface morphology

To determine the fractured surface properties of the developed hybrid composite with different compositions of hybrid filler loadings, the cross-section of fractured surfaces after tensile tests were carried out for the FESEM surface morphological analysis, as shown in Figure 7. A significant modification was observed due to MoS_2 filler loadings in equal wt.% with GnPs filler contents in the pure PA-6,6 matrix phase. Pure PA-6,6 matrix shows a flat and smooth fractured surface with ridges. The material pill-out (see Figure 7A) with a cave-like structure exhibiting the fracture dimple, suggesting the ductile fracture of the PA-6,6 polymer matrix. The pill-out fractured surface shows the micro-voids (see Figure 7A), which may be the probable cause of the ductile fracture. Further, the fractured surface of PGM hybrid composites demonstrates the smooth river and tongue-like pattern (see Figure 7B) due to the layer-wise structure of GnPs and MoS_2 filler contents in the hybrid composite system. Such structural patterns show the ravines-like structure with no fractured gully and the tearing-shaped pattern, suggesting that the hybrid composites more likely underwent the brittle fracture due to experiencing the higher tensile resistance and higher strain. Also, the typical wrapping structure exhibits better dispersion of hybrid fillers. The small red arrows show the even distribution of MoS_2 particles (see Figure 7B). At a fractured surface with 3 wt.% of hybrid fillers reinforcement (see Figure 7C), the ravines-like structure, as well as the penetration crack appears, suggesting the fractured gully-like morphology. This may be more likely because the composites experience fiber-like cracking and material pill-out. Also, from Figure 7C, the squamous-like morphology with a relatively rough structure was identified, suggesting the brittle behavior of the composites, which provides a reason for the decrease in toughness. The PGM hybrid composites containing the GnPs filler contents provide good compatibility with the PA-6,6 matrix interface that further leads to crack deflection and disproportionation, causing the absorption of tensile fracture energy and impeding crack propagation. As a result, the tensile strength is increased. The fractured surface morphological observations (see Figure 7D) for 5 wt.% of GnPs:MoS_2 reinforcement indicate the rough-surface morphology along with wrinkles and sheet faults with stacked sheet arrangements. This exhibits significant and more effective reinforcements that cause the toughening effect. Further, the MoS_2 loading causes interlaminar slippage among the MoS_2 stacked layers. Additionally, the MoS_2 filler loading promotes discontinuity of the PA-6,6 matrix phase, which leads to the formation of micro-cracks (indicated by a small arrow in light blue color in Figure 7D), which potentially behave like crack initiation and crack propagation zones of material failure. This agrees with the increased tensile strength property. The obvious possible agglomeration phenomenon seemed to occur at higher contents of hybrid filler loadings (at 10 wt.%), indicating inhomogeneous filler dispersion and relatively poor matrix-filler interfacial adhesion (see Figure 7E). This increases the poor retention or filler binding capability with the matrix PA-6,6 phase. The developed micro-cracks, wrinkles, and aggregation of hybrid fillers can be clearly observed on the fractured surface, which potentially causes premature material failure. The surface morphology shown in Figure 7E coincides with the result of tensile strength, which shows a significant reduction at 10 wt.% of filler reinforcement.

3.7 | Impact strength

The hybrid filler-based PGM composite with equal wt.% of GnPs and MoS_2 reinforcement for different compositions

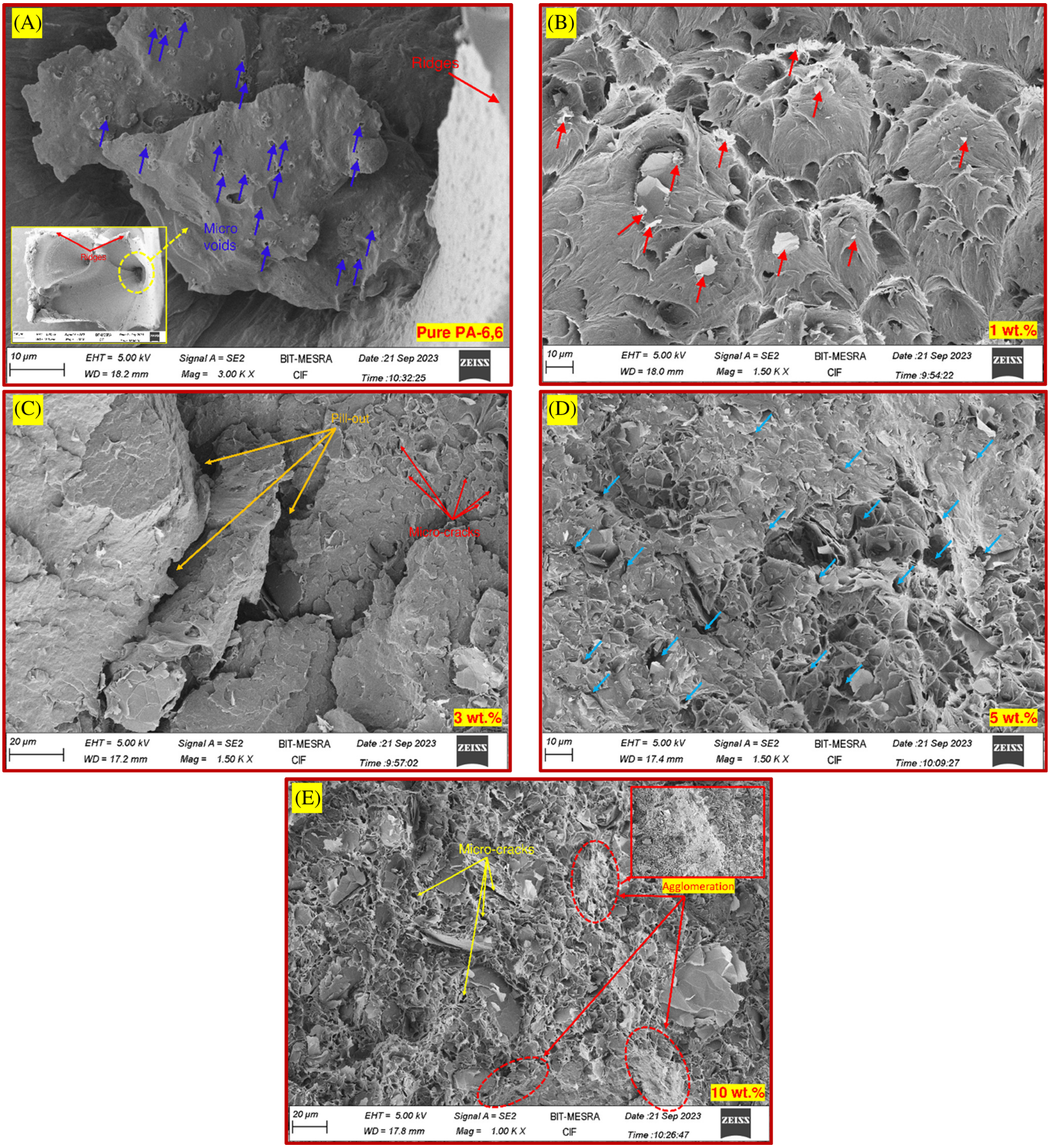

FIGURE 7 FESEM images showing the tensile fractured surfaces of (A) pure PA-6,6; (B) 1 wt.% PGM (C) 3 wt.% PGM-1; (D) 5 wt.% PGM, and (E) 10 wt.% PGM hybrid composites.

viz. 1, 3, 5, and 10 wt.% can be clearly delineated in Figure 8. From Figure 8, it can be observed that the pure PA-6,6 polymer matrix-based sample experiences the maximum impact strength of about 50.147 ± 1.463 J/m. Further, a definite falling trend in impact strength can be clearly seen in Figure 8. The reduced impact strength of the proposed composite system results in the synergistic action of GNP:MoS_2 hybrid fillers in equal wt.%. The high impact strength of unfilled PA-6,6 is related to the ductile nature possessed by the PA-6,6 polymer matrix. On the other hand, the addition of hybrid filler particles continually reduces the impact strength.

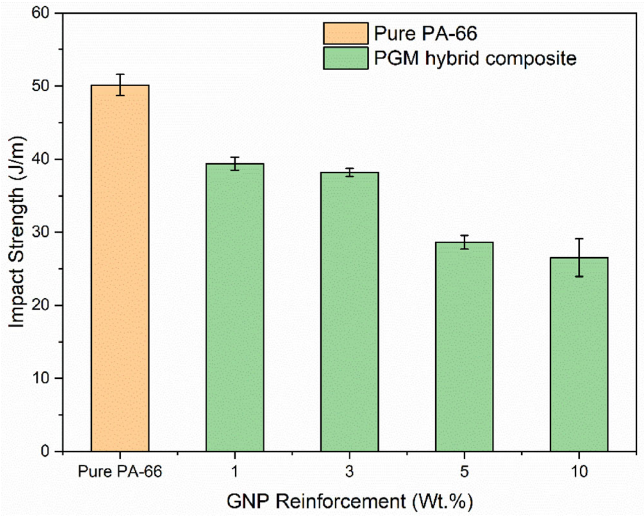

FIGURE 8 Impact strength of the hybrid PGM composite.

This drop in impact strength is attributed to the hybridized effect of hybrid filler reinforcement.[71] From Figure 8, it can be observed that at 1 and 3 wt.% of hybrid filler loading, no significant reduction was obtained, but at higher wt.% of hybrid filler addition, a significant reduction in impact strength of about was observed. At higher filler reinforcement, that is, at 5 and 10 wt.%, the reduction in impact strength of ~42% and ~47% than pure PA-6,6 was observed. This lower impact strength is consistent with prior research in the literature, which in that the increased loading concentration of graphene-based hybrid fillers reduced the impact toughness of thermoplastic.[2,72] This also established the reason for the lower impact strength, which is owing to the brittle fracture behavior of the PGM hybrid composite. In this investigation, there was no indication of plastic deformation of the PA-6,6 matrix. The collaborative impact of hybrid fillers weakens the composite structure due to the formed agglomeration of filler particles with increased wt.% of hybrid filler.[20,73,74] This may increase the possibility of induced void development and uneven filler mixing with the matrix phase. Furthermore, the agglomerated filler contents produce microcracks that function as internal notches, leading to sudden fractures from these notches, resulting in reduced impact strength.[20]

Furthermore, the phenomenon of decreasing impact strength with an increase in hybrid filler contents can be supported by the FESEM micrographs, as demonstrated in Figure 9. The fracture surface of pure PA-6,6 under sudden loading, was found to be flat and smooth, as shown in Figure 9A. At higher magnification, it can be identified as a cleavage fracture, which can be clearly visualized as a sudden burst of energy on the breakdown, indicating a relatively more ductile fracture mode. This fracture behavior consumes a large amount of energy.[75] Therefore, the pure PA-6,6 indicates the highest impact strength. Further, reinforcing the PA-6,6 matrix with hybrid filler contents exhibits ductile to brittle transition. Higher filler addition in the composite system will make the sample more brittle. At 5 wt.% of hybrid filler addition, the material becomes relatively more brittle than the pure PA-6,6 matrix and undergoes the low energy absorbed due to sudden loading, which is further found to have less plastic deformation.[76] This can be clearly seen in Figure 9B. At 5 wt.% of hybrid filler loading, a flat and rough surface with macroscopically brittle-like surface morphology. Broken particles were found to be accumulated on the surface, which can be clearly seen in Figure 9B. Thus, the notched impact strength of PGM-5% sharply decreased compared to pure PA-6,6. Similarly, reinforcing the hybrid fillers at higher wt.% resulted in a more brittle fracture due to sudden impact loading, significantly reducing the energy absorbed by the PGM-10% hybrid composite. The fractured samples showed numerous cleavages and micro-cracks, indicating a low energy absorption. This brittleness could be attributed to the agglomeration of filler particles at higher concentrations, leading to poor filler-matrix interaction and adhesion. The increased filler content creates stress concentration zones, where crack initiation occurs, promoting surface defects and ultimately reducing the composite's impact strength.[77]

4 | TRIBOLOGICAL PROPERTIES

4.1 | Friction and wear behavior

Figure 10A–F exhibits the evolution of the coefficient of friction during the wear test of the prepared PGM hybrid composites. Figure 9 shows the decreasing as well as an increasing trend in CoF during the experimental runs of wear test for varying wt.% of the PGM hybrid composites. Such trends in CoF settle to steady-state conditions after some time once the transfer films are formed on the counterface surface during the wear test mechanism. The produced transfer films during the tribology test were intact with the steel ball counterface surface, and the CoF reached the constant value.[78,79]

In the case of all polymers, the CoF initiated with an initial running-in behavior and ultimately transitioned into a stable, steady-state period.[80] Figure 10A–F shows the experimental runs performed at room temperature (RT) that exhibit stable friction behavior after some time of the test run-in period under all abrading frequencies and loads. Furthermore, Figure 10B,D,F exhibit the experimental runs with transient periods among the steady-state stages, showing the instabilities that

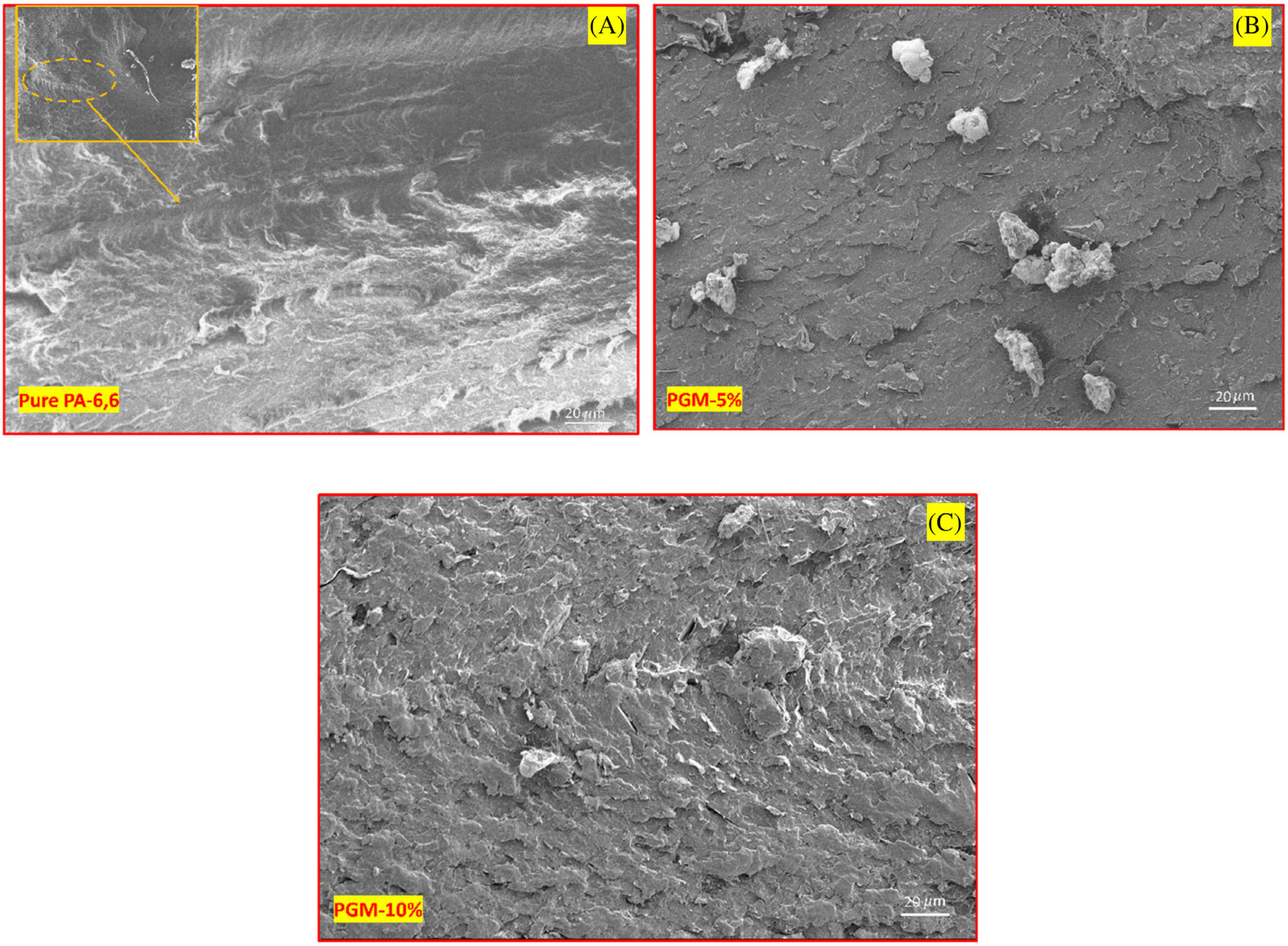

FIGURE 9 FESEM micrographs indicating the fracture surface morphology of impact test samples of (A) pure PA-6,6, (B) PGM-5%, and (C) PGM-10%.

occurred in CoF behavior. Such variations in the CoF cause an unpleasant squeaky noise at the interface, which is especially problematic in industrial applications. As a result, the tribo-pair experiences continuous wear due to the plowing of asperities from the steel ball counterface surface.

Moreover, from Figure 11A, a decreasing trend in CoF was observed, up to 5 wt.% of hybrid filler loading at a low abrading frequency under all normal loads, compared to Pure PA-6,6. Subsequently, an increasing effect in CoF was identified on higher filler loadings (i.e., 10 wt.%). A reduction in CoF of approximately 30%, 63%, and 72% compared to pure PA-6,6 was observed for 1, 3 and 5 wt.%, respectively. Similarly, a significant decreasing trend in CoF was observed for 30 and 50 N loads. Specifically, under a 30 N normal load, a reduction in CoF of approximately 24%, 37%, and 58%, and under a 50 N normal load, the reductions in CoF of approximately 17%, 21%, and 49% were noted at 1, 3, and 5 wt.% of hybrid filler addition, respectively. Furthermore, at higher hybrid filler loadings (at 10 wt.%), a significant increase in CoF was identified.

A similar trend at a 25 Hz frequency level was also identified under similar test conditions, as shown in Figure 11B. At high abrading frequency and 10 N load, a decrease in CoF was observed to be approximately 15%, 13% and 31% for 1, 3 and 5 wt.%, respectively. Similarly, for 30 and 50 N normal loads, the CoF was found to be reduced by approximately 24%, 25% and 38%, as well as 22%, 28% and 36% than pure PA-6,6, respectively, for 1, 3 and 5 wt.% of hybrid filler reinforcement. Despite these observations, at higher wt. % of hybrid filler loadings, the PGM composites indicated higher CoF. This result indicates a good agreement with the previous studies found in References 81,82.

Furthermore, apart from the excellent CoF behavior exhibited by the PGM hybrid composite, it can be noted that the introduction of MoS_2 filler contents in varying concentrations imparts a lubrication effect, leading to a decreased CoF. Each composition of hybrid filler loadings

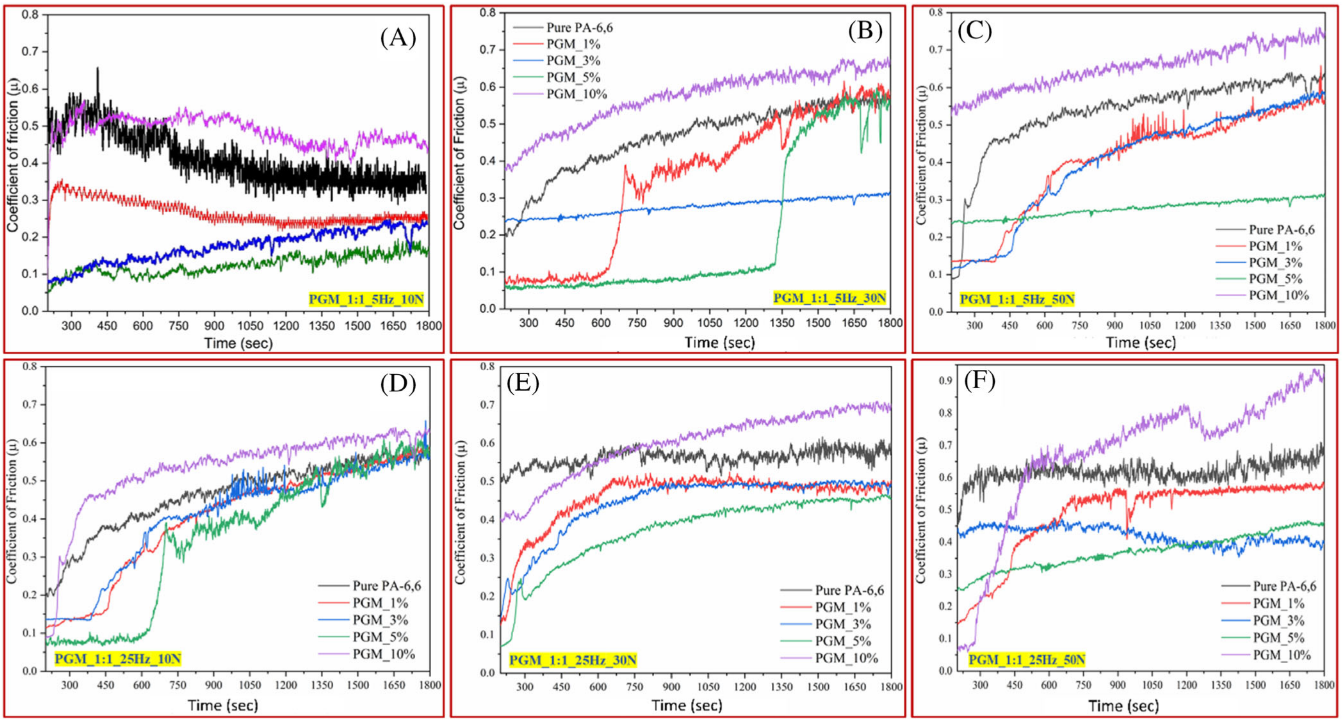

FIGURE 10 Evolution of CoF for pure PA-6,6 and its PGM hybrid composite.

indicated an increased CoF with an increased applied normal load and lower CoF with increased filler reinforcements up to 5 wt.%. In the sliding wear of polymers against a metallic counterface surface, friction is caused by the adhesion and abrasion mechanism, which is possibly due to the real contact area and the shear stress of the softer material.[83,84] The effect of load on friction and wear properties is primarily due to changes in the surface temperature. As the load increases, the accumulation of frictional heat on the surface also increases, leading to a higher temperature. This rise in temperature affects the viscoelastic property of the material, thereby enhancing its tribological performance.[84] Further, higher filler loading (say 10 wt.%) shows a significant increase in CoF. This is due to the improper filler-matrix interfacial adhesion and interaction as well as particle agglomeration. At a high abrading frequency, the CoF of an individual composition was found to be increased. The increasing normal load may be attributed to imparting a higher tangential force that acted at the contact of the sample and the steel ball counterface surface, resulting in a higher CoF.[45,85]

Furthermore, an improved wear behavior of the PGM hybrid composite was identified and can be seen in Figure 11C,D. From Figure 11C,D, a continuous decreasing trend in the specific wear rate up to 5 wt.% of hybrid filler loadings corresponding to low and high abrading frequency under all loads was observed. The acquired results are in the same order of magnitude as the results obtained by.[86] From Figure 11C, at a 5 Hz frequency and a 10 N normal load, the specific wear rate (W_{SR}) decreased by approximately 44%, 66%, and 87% for hybrid fillers at varying weight concentrations of 1%, 3%, and 5%, respectively, under low abrading frequency and 10 N load. Further, the W_{SR} was reduced by approximately 26%, 59%, and 84%, as well as by approximately 38%, 62%, and 85% compared to the pure PA-6,6 matrix for 30 and 50 N load, for 1, 3 and 5 wt.% of hybrid filler loadings. Similarly, the hybrid filler concentration at 5 wt.% demonstrated a significant decrease in wear behavior. At an abrading frequency of 25 Hz with 1, 3, and 5 wt.% of hybrid filler concentrations, as illustrated in Figure 11D, the W_{SR} decreased by approximately 14%, 66%, and 87% under a 10 N normal load. For a 30 N normal load, the reductions were approximately 14%, 69%, and 87%, respectively. Under a higher load of 50 N, the W_{SR} characteristics were reduced by approximately 17%, 56%, and 85%, demonstrating improved wear resistance characteristics compared to the pure PA-6,6 matrix. Therefore, the present study reveals that the PGM composite with 5 wt.% of hybrid filler loadings is the optimum amount of hybrid filler (GnPs:MoS_2) reinforcement that exhibits the highest wear resistance characteristics compared to pure PA-6,6 at all loads and all abrading frequencies.

Moreover, at a higher concentration of hybrid filler (10 wt.%), an increased behavior in CoF and W_{SR} was identified, as demonstrated in Figure 11C,D. Regardless of the amount of filler content, a positive effect in CoF

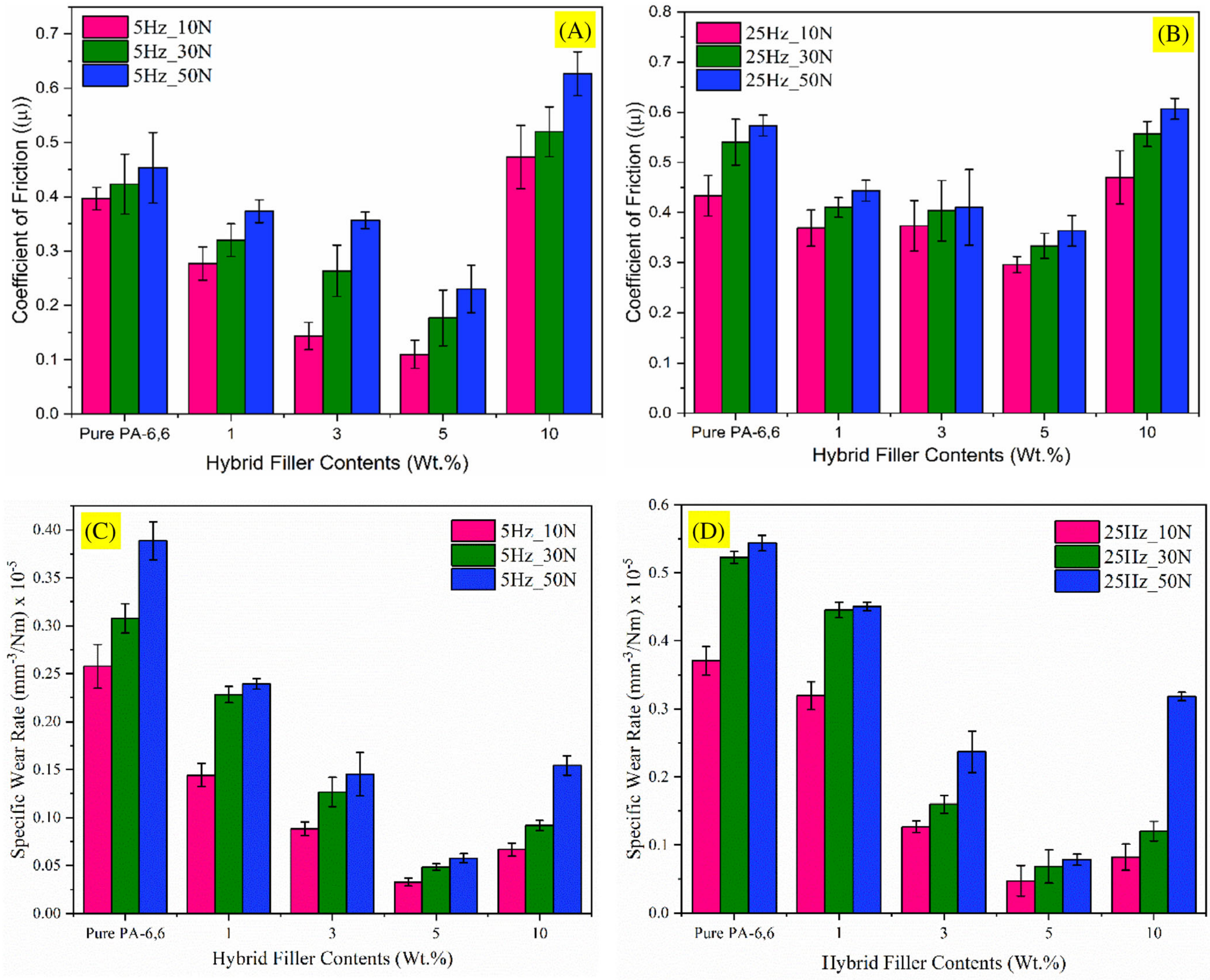

FIGURE 11 Effect of hybrid filler contents and normal loads on CoF and W_{SR} characteristics under dry sliding contact condition for PGM hybrid composite during wear test run under 5 and 25 Hz abrading frequency for (A, B) CoF and (C, D) W_{SR} characteristics.

and W_{SR} characteristics was observed in up to 5 wt.% of hybrid filler reinforcement. This improvement in CoF and W_{SR} is largely due to the role of MoS_2 as a solid lubricant that contributes to enhancing the wear resistance behavior of the PGM composite system. Therefore, it can be stated that the wear resistance characteristics and the reduced friction coefficient are attributed to the high hardness and enhanced tensile strength of the hybrid composite system.[85] The reduction in wear rate characteristics of the PGM hybrid composites compared to the pure PA-6,6 polymer matrix may be attributed to the strong interfacial interaction between the filler and matrix, as well as the protective layer formation by the filler particles over the entire volume of the matrix which also has been reported in the previous studies.[85,87] Additionally, in the case of MoS_2 particles, oxidation reactions contribute to improved adhesion between the counterface and the transfer film, enhancing wear properties. These findings support the notion that incorporating small quantities of these fillers is beneficial for reducing wear characteristics, as previously documented in the literature.[88]

The reciprocating abrading frequency is a significant parameter in the friction and wear behavior of the polymer composite system. As the frequency increases, W_{SR} characteristics of the pure PA-6,6 and its PGM hybrid composites were observed to increase due to high frictional heat generation during the abrasion with increasing normal loads. At high abrading frequency and high normal load, the material extrudes out from the worn surfaces due to the softening of the material, and a melt blend accumulates over the worn surface, leading to the formation of a transfer film layer on the steel ball counterface surface.[89] These transfer film layers reduce the scraping effect of the counterface, protect the worn surface from direct rubbing by the counterface surface, and stop the further removal of the material from the worn surface.[85,88,90,91] This is owing to the solid lubricating and shearing effect of the filler contents, which results in a considerable decrease in friction and wear when compared to unfilled polymer materials.[92]

The frictional heat generation results from the deformation of material (due to plowing action) that occurs during the wear test at the actual contact point, leading to the formation and breakdown of adhesive bonds.[93] The frictional heat also increases the temperature at the contact interface of the substrate and counterface, which elevates the temperature to the softening temperature of the material. Consequently, the stiffness characteristics of the material were found to be reduced, and the real contact area between the substrate plate and counterface surface contact increased, resulting in a corresponding increase in CoF.[94] Additionally, at high abrading frequency with increased applied load, the large real contact area generates an increased number of adhesive contact points, which requires more energy to overcome from these increased number of adhesive bonds, which in turn gives rise to the CoF of the material. Previously, it has been reported that the elevated temperature at the contact zone is a crucial factor influencing the friction behavior of the material.[95,96] Further, the higher CoF resulting from an increased temperature at the contact zone has also been previously reported.[93,97–99] A similar phenomenon of friction and wear behavior were also identified in the present study. Therefore, it can be concluded that the PGM hybrid composites containing the hybrid reinforcement of GnPs and the MoS_2 impart a positive effect on the friction and wear behavior of the prepared hybrid composites.

4.2 | Worn surface morphological analysis

In Figure 12A–C, the representative FESEM images illustrate the worn surface morphology of PA-6,6 and its PGM hybrid composites for better comprehension of the wear mechanism. From Figure 12A, the wear mechanism of pure PA-6,6 matrix under both abrading frequencies for all loads can be clearly seen. At low abrading frequency (5 Hz) and low load (30 N), the surface was smoother, occurring micro plowing with minimal wear, minimal debris and mild abrasion on the worn surface, resulting in low friction and wear rate. With increasing loads (50 N) and high abrading frequency (25 Hz), a clear borrow-pit (or dent) with higher abrasive wear, cracks and micro-cracks, along with the larger debris, can be clearly seen.

Similarly, Figure 12B delineates the worn surface morphology under a low abrading frequency (5 Hz) at all normal loads for PGM composite with 5 and 10 wt.% of hybrid fillers. The FESEM images shown in Figure 11B evidenced that the 5 wt.% of hybrid filler contents exhibited minimum wear pure PA-6,6. However, the increasing load from 30 to 50 N exhibited increased wear due to developed micro-cracks (shown in the black arrow) and material pull-out. Additionally, the role of the hybrid filler content in the wear rate is evident, as at higher wt.% (10 wt.%) of hybrid fillers, the wear rate increases, showing high abrasive wear with broken material, material pullout and wear debris accumulated on the worn surface. The entire surface was inherited from the micro-cracks developed during the sliding action, which may be attributed to the higher filler contents over the matrix phase.

Similarly, the worn surface morphologies under high abrading frequency (25 Hz), at higher reinforcement (5 and 10 wt.%) of hybrid fillers for 30 and 50 N applied load, can be clearly seen in Figure 12C. A significant improvement in wear behavior was observed at 5 wt.% of hybrid filler addition, and minimum surface damage with minimal abrasives and material pullout with fine wear debris compared to pure PA-6,6 was observed on the worn surface. However, 10 wt.% of the hybrid filler loadings further increased the surface damage with high abrasives accumulated on the worn surface along with broken particles. This may be due to the excess amount of hybrid filler contents, which further develop rough, exfoliated, and discontinuous worn surfaces under high abrading frequency and high load (50 N), which can be clearly seen in Figure 12C. The exfoliated and unevenly worn surface with excessive melt materials extruded out and accumulated over the worn surface after detachment from the counterface surface during the sliding contact action. Excessive wear debris, broken and pull-out material can be seen, resulting in a higher COF and wear rate. Further, a comparatively lower load (30 N) indicated less surface damage with micro-cracks formation and micro-plowing action, indicated a mild abrasive wear phenomenon. The encrusted type morphology with surface cracks can be clearly identified. However, nano or micro fillers in polymeric matrices hinder crack propagation during sliding by transferring stress efficiently from the polymer matrix to the nano or micro filler contents, resulting in a milder wear mechanism.[100] Therefore, It can be further concluded that a significant improvement in the wear behavior of the PGM hybrid composite was observed to be at 5 wt.% of hybrid filler reinforcement. These enhanced wear-resistant characteristics of the developed novel hybrid composite demonstrate that the normal stress

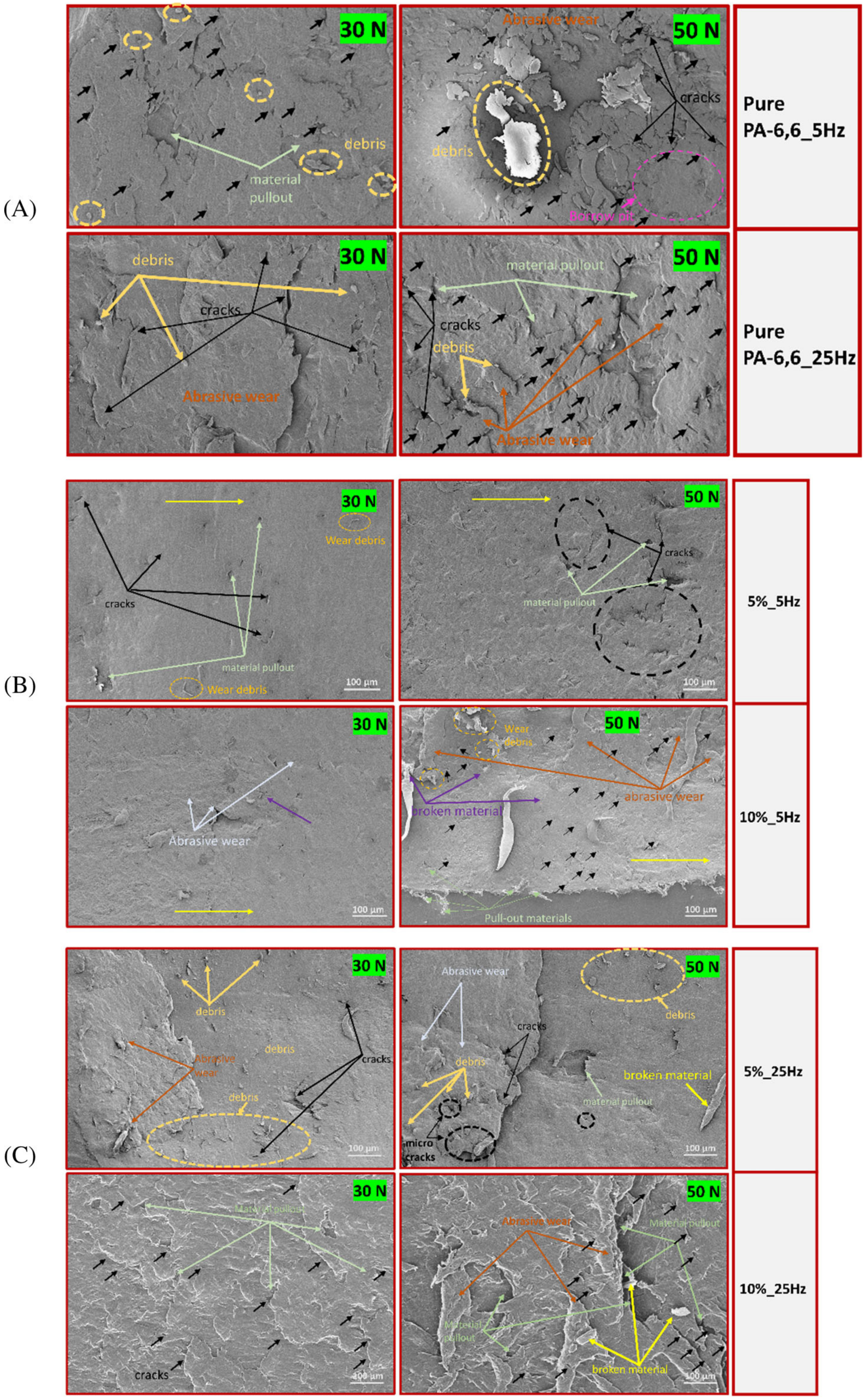

FIGURE 12 FESEM micrographs for the worn-out surface of (A) pure PA-6,6 under both abrading frequencies, (B) PGM composite under 5 Hz and (C) PGM composite under 25 Hz abrading frequency.

due to the applied normal loads during the indentation of the steel ball counterface was effectively carried out by hybrid fillers and then transferred to the PA-6,6 matrix phase.[101] Thus, the developed hybrid composites filled with 5 wt.% hybrid filler reinforcement provided a significant contribution towards the improvement of their tribological performance.

4.3 | Formation of transfer film

Incorporating solid lubricants such as MoS_2 along with GnPs as hybrid fillers into a polymer matrix is an effective way to influence the tribological and wear behavior of polymer hybrid composites, as well as provides a significant reduction in CoF and wear rate.[102–104] In the process

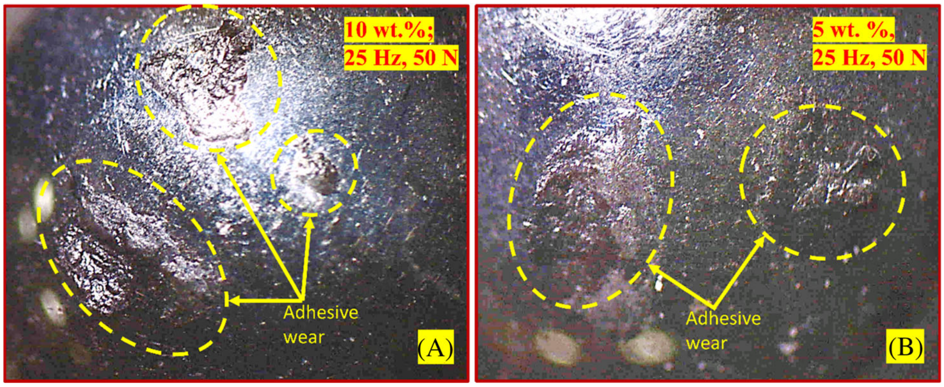

FIGURE 13 Optical microscopy images of steel ball counterface surface post-tribology experiment indicating the formation of transfer film of (A) 10 wt.%, under 25 Hz, 50 N (B) 5 wt.%, under 25 Hz, 50 N.

of sliding surfaces over one another, if the interfacial adhesion between asperities of the tribo-pair exceeds the cohesive strength of the softer material, commonly polymers, the material starts getting transferred to the counterface (like a steel ball or steel pin) and leads to the formation of a thin transfer film.[78,80] The formation of such a layer of a transfer film over the surface of the steel ball counterface was also observed in our investigation, as shown in Figure 13A,B, which are confirmed by the optical images of the steel ball counterface by optical microscope.

The captured optical images of the PGM hybrid composites samples post tribology test indicate the formation of transfer film on the steel ball counterface surface during the long wear test run-in period under high abrading frequency (25 Hz) and higher normal load (30 & 50 N). Such formation of transfer film protects the composite surfaces from further removal of the softer polymer material due to the effect of plowing action against the harder asperities of the steel ball, resulting in an occurrence of adhesive wear leading to the formation of a thick and lumpy transfer film as shown in Figure 13A. From Figure 13B, clear deposited patches of non-continuous transfer film can be clearly seen.

Such non-continuous behavior of the deposited softer material exhibits the improper adhesion characteristics of the transfer film over the steel ball counterface, which further signifies the high wear rate of the developed hybrid composites. This behavior of hybrid composites may be due to the loading of higher wt.% of GnPs:MoS_2 hybrid filler contents of the developed hybrid composites. Further, the transfer films formed during the wear test run-in period on the steel ball counterface (see Figure 13B) exhibit a thin and tenacious film with lower wt.% of hybrid filler contents (i.e., 5 wt.%), which indicates an improper full coverage over the contact area of the steel ball counterface. The thick and lumpy structure of a non-continuous transfer film indicates the higher contents (i.e.,10 wt.%) of the hybrid filler reinforcement, which shows the thick, uniform and non-continuous film surface on the contact of the steel ball counterface surface, leading to the high wear of the hybrid composite. At higher filler concentrations, this non-continuous transfer film may be due to the presence of higher content of MoS_2 filler particles. This suggests that an increased concentration of MoS_2 loading results in the development of a substantial transfer film on the steel ball counterface surface. However, this transfer film tends to be of poor quality and fragile, ultimately leading to an elevated wear rate.

5 | CONCLUSION

The study successfully fabricated a hybrid composite using PA-6,6 matrix reinforced with hybrid fillers (GnPs:MoS_2) through a melt-mixing and injection molding technique. The current study focused on the effects of hybrid fillers on the structural, thermal, mechanical, and tribological properties of the developed novel PGM hybrid composite. By incorporating the fillers in equal concentrations, the study aimed to analyze the combined effects of these materials on the composite's ability to withstand high temperatures, sustain heavy loads, and perform under high-frequency friction and wear conditions. The results exhibited that hybrid fillers (GnPs:MoS_2) at 5 wt.% reinforcement significantly improved the thermal, mechanical and tribological properties. The key findings of the present research work are summarized below:

1. The increased thermal stability was identified with a significant temperature rise of ~116°C (i.e., ~26%) than pure PA-6,6 at 5 wt.% of hybrid filler addition. This was confirmed through the effective filler dispersion and demonstrated by the XRD pattern study. Further, DSC thermograms also identified a significantly improved crystallization temperature.
2. An improved tensile strength and modulus properties were identified at increased hybrid filler concentrations. The tensile strength property was found to be enhanced by ~90%, and the modulus was observed to be increased by ~29% due to hybrid filler loadings. Additionally, the fractured surface morphologies

provided a better understanding and deep insights regarding the failure mechanism of the composites. However, a decreasing trend in impact behavior was also observed with increasing concentration of hybrid filler.

3. Furthermore, the study indicated improved friction and wear behavior of the developed hybrid composites, indicating that the solid lubricant MoS_2 had a distinct influence on friction and wear behavior. A continuously decreasing trend in CoF Behavior was identified with increased hybrid filler contents. The optimal CoF was found at 5 wt.% of hybrid filler addition than unfilled PA-6,6. Additionally, with increasing load under both abrading frequencies (5 and 25 Hz), the increased CoF was identified for each composition.
4. Increased load and high abrading frequencies had a pronounced effect on the friction and wear of the unfilled PA-6,6 and its PGM composites. The enhanced wear resistance characteristics were also observed at 5 wt.% of hybrid filler reinforcement under all loads and abrading frequencies compared to pure PA-6,6. The results also demonstrated the increased wear rate under all abrading frequencies as the loads were elevated for each composition of PGM composites.
5. The worn surface morphologies clearly impart positive insights about the improved wear characteristics with abrasive and adhesive wear mechanisms.

Therefore, the present novel work and investigation confirm that 5 wt.% of hybrid filler (GnPs:MoS_2) reinforcement was identified to be an optimal concentration that produces an effective thermal, mechanical and tribological performance of the developed hybrid PGM composite.

Due to its enhanced thermal, mechanical, and self-lubricating properties, the developed novel composite may have potential applications in various components like gears, bushing, and under-the-hood parts. Lightweight structural components requiring wear-resisting properties, as well as bearings, seals, and wear plates that operate under high load and friction conditions, could have the potential benefit from improved wear resistance and friction characteristics of such hybrid composites. Additionally, the combined effect of GnPs and MoS_2 can be further investigated for their electrochemical performance and could be an alternative to be utilized as an electrode material for supercapacitors.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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